

DATABASE DESIGN PROJECT 2019-20
FLIGHT BOOKING SYSTEM

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A. Student Details

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Student number: R00170779

Class: SDH2C

B.project description

This project is to implement an application system related to a flight booking system. The projects includes both the database and front-end (Java) aspect to it. The database is done with MySQL workbench and the front-end is done with eclipse. The project focuses on the general purpose procedural programming language.

Detailed description and specifications:

There is an option for both the customer and the staff.

The customer should be able to log-in or sign up. A customer not existing in the database will be given a randomly generated customer ID after signing up. I am assuming that every customer enters unique username and password.

A customer can make many bookings. He/she provides his/her customerID before booking process begins. The bookingID, luggageID and paymentID is automatically generated and displayed to the customer.

There are only three flight classes(Economy, Business or First)

There are only two luggage types(Carry-on or Check-in)

There are only three modes of payments(Visa Debit Card, Visa Credit Card and PayPal)

A customer also has the option of viewing his/her bookings, since there are customers with multiple bookings.

Customers can also cancel a booking, but they must provide their bookingID.

The staff also has an option to add or remove flights. For simplicity, I am allowing the staff to only add flights and not the airports.

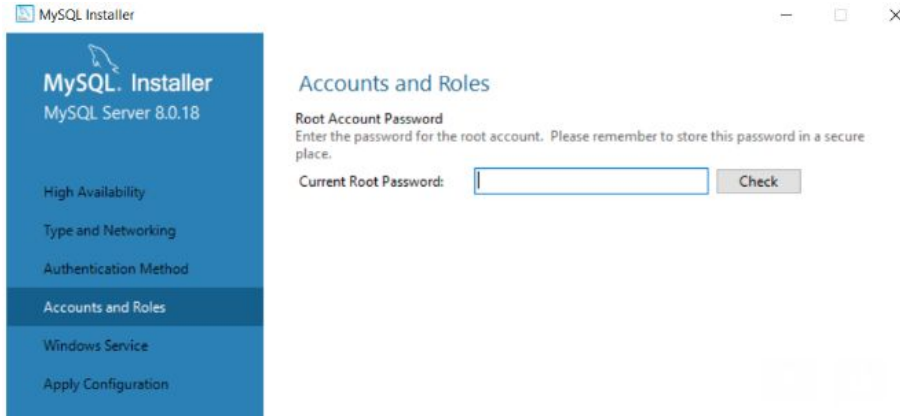
Any price, totals, or amounts are in decimals rounded to two decimal places.

The date format is in yyyy-mm-dd

C. technical issues dealt with

7/11/2019

I tried to re-install mysql workbench. I had problem of connecting to it. I had previously created a root account last year, but I forgot the password. I tried to reinstall Mysql(The whole thing) and this came:



It kept asking for my previous account's password, which I did not remember, even after uninstalling the mysql.

I then tried to manually 'deep-clean' mysql from windows, as there would have been some 'leftovers' which would have failed to uninstall in my system.

After that, I was able to reset my password and create a new account.

Accounts and Roles

Root Account Password
Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:
Repeat Password:
Password strength: **Medium**

MySQL User Accounts

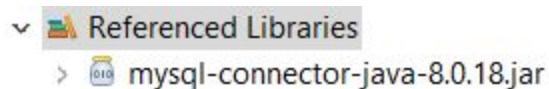
Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.

MySQL User Name	Host	User Role	
			Add Edit Delete

18/11/2019

I had to install the mysql-connector-java-8.0.18.jar(Referenced Libraries).

Problem: The JDBC already had a Referenced Library version of 6.0. I soon realized that I needed a library that is of the latest version and the previous one was for the wrong version.



22/11/2019

I had problems converting my Visual Paradigm ER diagram to mysql. I did it through the 'Generate Database' and 'Configure DB' in the VP. I finally got it, after trying for almost 3 hours.

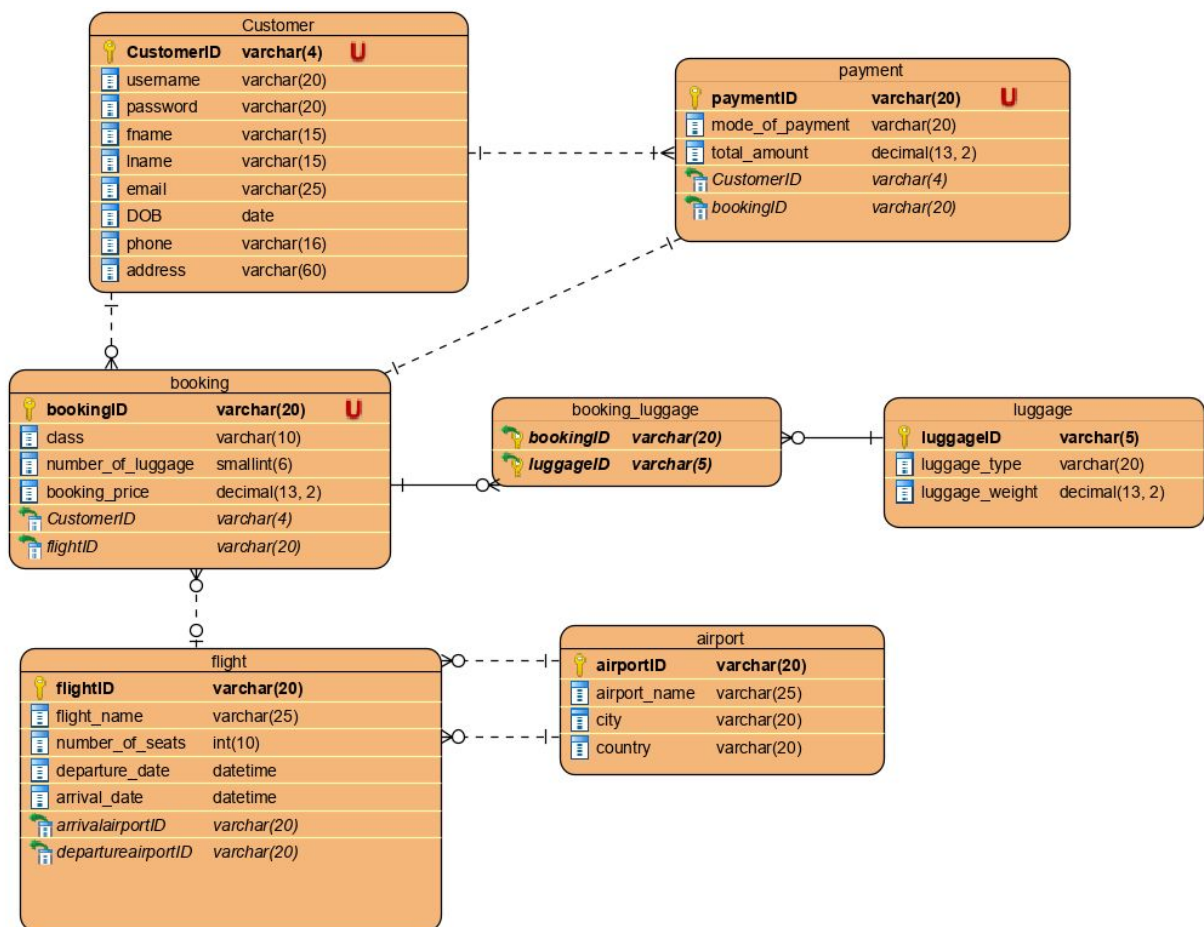
It seems one of the data I had given as an input was incorrect. After trying to figure out which I got the connection to be successful. This is the correct one:

The screenshot shows the 'Database Generation' dialog box. The 'Output Path' is set to 'C:\Users\samyu\Desktop'. The 'Generate Database' dropdown is set to 'Drop and Create Database'. The 'Schema' dropdown is set to '<All>'. The 'Export to database' checkbox is checked. The 'Generate DDL' and 'Generate Comment' checkboxes are also checked. The 'Upper Case SQL' checkbox is checked. The 'Formatted SQL' checkbox is checked. The 'Custom DDL' button is visible. The 'Generate Individual DDL' checkbox is unchecked. The 'Separate Create/Drop DDL' checkbox is unchecked. The 'Generate Sample Data' dropdown is set to 'Yes (With Auto Generated PK)'. The 'Quote SQL Identifier' dropdown is set to 'Default(Auto)'. The 'Column Order' dropdown is set to 'Normal'. The 'Table Charset' dropdown is empty. The 'DDL Extension' dropdown is set to '.ddl'. The 'Connection Provider Class' dropdown is set to 'bernate.service.jdbc.connections.internal.C3P0ConnectionProvider'. The 'Connection' dropdown is set to 'JDBC'. The 'JDBC' section is expanded, showing 'Connection Pool Options' and 'Database Configuration'. The 'Driver' dropdown is set to 'MySQL5 (Connector/J Driver)'. The 'Driver file' is set to 'ds\mysql-connector-java-8.0.18\mysql-connector-java-8.0.18.jar'. The 'Connection URL' is set to 'jdbc:mysql://127.0.0.1:3306/r00170779'. The 'User' is set to 'root' and the 'Password' is masked with dots. The 'Test Connection' button is visible. The 'OK', 'Cancel', and 'Help' buttons are at the bottom.

The screenshot shows the 'Database Configuration' dialog box. The 'Language' dropdown is set to 'Java'. The 'MySQL' checkbox is checked. The 'MariaDB', 'MS SQL Server', 'Oracle', 'HSQL', 'Sybase ASE', 'Sybase SQL Anywhere', 'PostgreSQL', 'Cloudscape/Derby', 'DB2', 'Ingres', 'OpenEdge', 'Informix', 'Firebird', 'FrontBase', and 'Cache' checkboxes are unchecked. The 'Database Setting' section is expanded, showing 'Version' set to '5.0.5 or higher', 'Driver' set to 'MySQL5 (Connector/J Driver)', 'Driver file' set to 'mysql-connector-java-8.0.18\mysql-connector-java-8.0.18.jar', and 'Connection URL' set to 'jdbc:mysql://127.0.0.1:3306/r00170779'. The 'User' is set to 'root' and the 'Password' is masked with dots. The 'Engine' section has 'Default' selected. The 'Test Connection' button is visible. The 'Database Driver Description' section is collapsed.

D. Database design

The snippet from visual paradigm,



E. Application Design.

<pre> FLIGHT BOOKING SYSTEM CUSTOMER ----- 1.SIGN UP 2.CUSTOMER LOG IN 3.BOOK A FLIGHT TICKET 4.VIEW A BOOKING 5.CANCEL A FLIGHT TICKET FLIGHT ADMINISTRATOR ----- 6.Add a flight: 7.Remove a flight: 8. QUIT PROGRAM Error handling: Enter choice: 9 Choices should be [1,2,3,4,5,6,7,8] please! </pre>	This is like a menu-driven system. Options 1-5 are for the customer. While 6 and 7 are for the flight administrator. 8 is common for both who wish to exit. If entered a number not 1-8 then it will display the menu again and handle the error. If anything else they have to restart.
<pre> Customer(SIGN UP) Enter choice: 1 CustomerID is C019 Enter your USERNAME: newcustomer11 Enter your PASSWORD: securepassword Enter your first name: samyu Enter your surname: tardis Enter your email: samyu.tardis@gmail.com Enter your DOB (yyyy-mm-dd): 1999-12-20 Enter phone no: 0857600984 Enter your Address: lids road, base town, cork, ireland </pre>	This is for a new customer. I auto generated a random CustomerID. The customer should follow the formatting (Uppercase, dates) etc. Eclipse: SQLQueryPrepared.java
<pre> Customer(LOGIN) Enter choice: 2 Enter username: newcustomer11 Enter password: securepassword You already have an account Your customerNo is: C019 </pre>	This is for an existing customer. Where he enters his/her username and password. If it exists, then it will display the customerID for their reference.

Customer(BOOKING) *image number 1. Given in the end of report.	Just like customerID. I will auto generate the bookingID, luggageID and paymentID. I have displayed the flight details and the customer will enter the flightID(following the formatting). He/she will enter the required details for each and a message of 'BOOKING COMPLETE', will be displayed.
Customer(View a booking) <pre> Enter choice: 4 Enter your CustomerID: C019 All your bookings are : 1. bookingID: B0014 class: Economy number_of_luggage: 1 booking price: 5.00 Customer_id: C019 flightID: TR9909 </pre>	If a customer has more than 1 booking(like customer C003), it will be convenient to display his/her booking details. This option is also useful if he/she decides to cancel a booking. They have a quick overview of their booking details.
CUSTOMER(CANCEL A FLIGHT TICKET) <pre> Enter choice: 5 Enter your bookingID: B0014 The Number or records deleted in payment is 1 The Number or records deleted in booking_luggage is 1 The Number or records deleted in booking is 1 Your booking has been cancelled </pre>	In order to delete a booking, the customer just has to provide his/her bookingID and automatically it will be deleted in the other tables. Eclipse: DeleteRecord.java
FLIGHT ADMIN(Add a flight) Image number 2	Here, I am assuming that there is only 1 staff that can add and remove flights. That is why I did not ask for login details. The staff will enter the details of the flight. The list of arrival airports will be displayed and the staff will enter the airportID(eg.A1) This is similar to choose the departure airports.
FLIGHT ADMIN(remove a flight) Image number 3	Similarly, here I display all the available flights, including the one added and the staff just has to enter the flightID to be deleted.

QUIT PROGRAM Enter choice: 8 Thank you for visiting out sight!	This is common to both the customer and the staff. After quitting, the system will give a message.
--	---

I have used prepared statements, insert, select, simple query and many more in the RunDB.java also.

Examples of the java code:

1. Adding new customer to the database by calling another class
RunDB.java

```
SQLQueryPrepared sqlQueryPrepared = new SQLQueryPrepared();
sqlQueryPrepared.queryStatement(con);
break;
```

- 2.

Using the DeleteRecord.java to delete the booking for customer:

```
DeleteRecord dr = new DeleteRecord();
dr.delete(con);
break;
```

3. viewing the booking details in RunDB.java

```
//view booking details
ResultSet rs2 = s.executeQuery ("Select * From booking Where customerID = " + "" + customer_id + "");

String viewbooking = "";
int booking_counter = 0;
System.out.println("\nAll your bookings are : ");
while (rs2.next ())
{
    String bookingNoVal = rs2.getString ("bookingID");
    viewbooking += bookingNoVal;
    System.out.print((booking_counter+1) + ". ");
    System.out.println("bookingID: "+bookingNoVal);
    booking_counter++;

    String bookclass = rs2.getString("class");
    System.out.println("class: "+ bookclass);

    String no_of_lugg = rs2.getString("number_of_luggage");
    System.out.println("number_of_luggage: "+ no_of_lugg);

    String book_prc = rs2.getString("booking_price");
    System.out.println("booking price:  "+ book_prc);

    String view_cust = rs2.getString("CustomerID");
    System.out.println("Customer_id:  "+ view_cust);

    String view_flight = rs2.getString("flightID");
    System.out.println("flightID:  "+ view_flight);

    System.out.println();
}
```

Inserting record in the luggage table in RunDB.java

```
Statement insertStmt1 = con.createStatement();

String insertSQL1 = " Insert into luggage(luggageID, luggage_type, luggage_weight) values ('"+luggageid+"', '"+luggage_type+"', '"+luggage_weight+"')";
int res1 = insertStmt1.executeUpdate(insertSQL1);
System.out.println("The Number of records inserted luggage is      " +res1);
// You May need to uncomment if Autocommit is not set
//con.commit();
insertStmt1.close();
```

Simple query in RunDB.java for customer login:

```
//Simple Query
Scanner sim = new Scanner(System.in);
System.out.println("Enter username: ");
String login_username = sim.nextLine();

Scanner sim1 = new Scanner(System.in);
System.out.println("Enter password: ");
String login_password = sim1.nextLine();

ResultSet rs = s.executeQuery ("SELECT * FROM customer WHERE username = " + "'" + login_username + "'" + "AND password = " + "'" + login_password + "'");
boolean existFlag = false;
String cust_no = "";
```

SQLQuery.java for getting the details of the airport table:

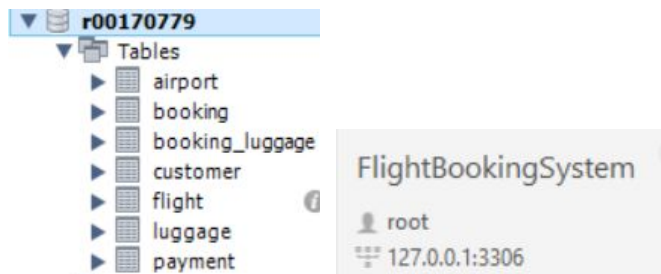
```
public void queryStatement(Connection con) {
    try {
        Statement stmt = con.createStatement();
        String qry = "select airportID, airport_name, city, country from airport";
        ResultSet rs = stmt.executeQuery(qry);

        while( rs.next()) {
            System.out.print("\nAirportID: " + rs.getString(1));
            System.out.print("\nAirport name: " + rs.getString(2));
            System.out.print("\nCity: " + rs.getString(3));
            System.out.print("\nCountry" + rs.getString(4));

            System.out.println("");
        }
        rs.close();
        stmt.close();
    } catch (Exception io) {
        System.out.println("error"+io);
    }
}

//query
```

Note:



```
Connection con = DriverManager.getConnection("jdbc:mysql://127.0.0.1:3306/r00170779?user=root&password=samrosh123" );
```

F. Plan/log

October:

14th: Plan to come up with a design for my DB using Visual Paradigm. Goal: 6-8 tables.

Get it approved with the lab professor. Planning on updating my documentation every lab.

18th: Worked and finished on the table inputs. I have to establish the connection between each table.

20th: Showed my table to professor. Turns out there is a bridging table involved. Have to work on that. Also was given suggestion to create a arrivalairportID and a departureairportID and establish it as a FK to the flight table.

28th: I finished the suggestions and established the relation between each table. Got them approved. Planning on inserting data.

November:

3rd: I inputted data into the table using VP. It automatically checked the data type and if I exceeded the count. I don't know if I have to manually generate the code to SQL or if it would be automatic.

7th: I tried reinstalling Mysql since I forgot my old root password. Could not establish connection. Had to do a deep cleaning. Got it working in the end after help from lecturer.

14th: Talked to lecturer after class, turns out, I have to create the project from a customer's point of view. Also make sure there is an option for a staff to add and remove a flight.

18th: I am going to use eclipse for my project. I am also going to use the JavaMySQL code as a reference when creating my application. I had a problem with my Reference library. Took the whole lab time to figure the problem.

22nd: I came to a conclusion that finished my DB and it was time to move on. Planning on converting from VP to MySQL. Update, Turns out it is hard. Could not figure it out during the lab. Stayed up most of the night and finally got it working.

23rd: Worked on my console application. Trying to figure out how prepared statements work. Having trouble with booking a flight for a customer. Planning to continue it tomorrow.

24th: Worked on my console application. Trying to use as many classes as possible to make it easier. Stayed up most of the night.

25th: Did my lab demo and planning to work on the feedback and comments. I should also include any changes I make to the documentation.

27th: Worked on my documentation and improving it. Trying to get the code for the insert statements directly from VP. I think I have to manually so it.

28th: Deadline for the project. Planning on printing the documentation and giving it during the lecture.

G. Conclusion/ Review

This project enabled me to use, implement and understand the concept of JDBC. I have a better understanding on how to connect both MySQL and java.

This project also made me realize that there can be technical errors in many forms and how there is always a way to find a solution.

It also encouraged me to keep and update a log, which is crucial nowadays in many aspects related to programming and progress. This project was beneficial in making me utilize the tools in Visual Paradigm which I did not know before. This project also focused on documentation which I feel is very useful. This project also encouraged me to utilize resources and information from the practise labs and lectures through which I was able to learn even more.

I am very grateful as this project taught me many beneficial and important lessons, which I can implement later on in my career/ studies.

H. References

I used the java code that was provided. JavaMySQL. From the labs @author ingres

Converting Vp to SQL:

https://www.visual-paradigm.com/support/documents/vpuserguide/3563/3584/85403_databaseconf.html

Fully uninstalling Mysql(performing deep copy):

<https://www.askvg.com/how-to-completely-uninstall-remove-a-software-program-in-windows-without-using-3rd-party-software/#:~:targetText=Open%20Control%20Panel%2C%20double%2Dclick,and%20click%20on%20Uninstall%20button.>

Installation process of Mysql:

<https://www.youtube.com/watch?v=WuBcTJnluzo>

Sample flight web page reference:

https://www.google.com/search?q=ryanair&rlz=1C1CHBF_enIE813IE813&oq=ryanair&aqs=chrome..69i57j69i60l3.1140j0j7&sourceid=chrome&ie=UTF-8

I. CHANGES MADE SINCE DEMO FROM LAB

I did my demo on 25/11/2019. Since then, I have changed and improved my JDBC code based on the feedback. I was told to include this.

These are the changes I made:

1. Old: A customer could only login with the username.

Problem: This was a security issue, a customer could get the username of another customer and view his/her details.

Change I made: I made sure that the customer could log-in with the username and the password. If the customer's account exists, it will show that and show his/her customerID.

```
Enter choice:
```

```
2
```

```
Enter username:
```

```
rose11tyler
```

```
Enter password:
```

```
badwolf10
```

```
You already have an account
```

Snippet of console: Your customerNo is: C001

2. Old: While signing up a new customer, the customer had to generate the customerID by themselves.

Problem: The customer could give a customerID already belonging to another customer.

Solution: I improved the code to randomly auto generate a customerID for the new customer.

Snippet of console:

```
Enter choice:
1
CustomerID is C010
Enter your USERNAME:
```

Snippet from the SQLQueryPrepared.java:

```
//auto generate customerID
Random cust_rand = new Random();
int c_no = cust_rand.nextInt(20);
String customerID = "";
while(c_no < 10) {
    c_no = cust_rand.nextInt(20);
}
System.out.println("CustomerID is C0"+ c_no);
customerID = "C0"+ c_no;
```

3. Old: For a customer to cancel his/her booking, I asked them to Manually enter their luggageID and then deleted from the luggage table and cancel it from the luggage table. Then I asked them to enter their PaymentID and cancel it from the payment table, And only then enter the bookingID to cancel it from the booking table. I did not use the bridging table to delete information.

Problem: This was very cumbersome to do in the console as it is not very customer friendly.

Changes I made: I made sure the customer only entered the bookingID.

Through that I could automatically delete the needed details from the payments and booking_luggage table. This was very user friendly and a huge improvement. The code can be found in the DeleteRecord.java.

I also automatically displayed in the

4. Old: My booking a flight option was not very clean. The user manually had to enter the dates, destination, arrival etc. and then it would display if the flights are available or not.

Changes I made: I made sure that the list of already available flights are shown so that it would be easier for the customer to choose.

Snippet from the console:

```

Enter choice:
3
Enter your CustomerID:
C001
1. flightID: FR6606
flight name: Airway
departure time: 2018-10-02 00:00:00
arrival time: 2018-10-02 00:00:00
departure airport: Cork
2. flightID: FR9896
flight name: Ryanair
departure time: 2019-12-01 00:00:00
arrival time: 2019-12-01 00:00:00
departure airport: Cork
3. flightID: FR1209
flight name: Ryanair
departure time: 2019-01-01 00:00:00
arrival time: 2019-01-01 00:00:00
departure airport: Heathrow
4. flightID: TR9909
flight name: Airlingus
departure time: 2018-12-21 00:00:00
arrival time: 2018-12-22 00:00:00
departure airport: Chennai International
5. flightID: ST8909
flight name: British Airways
departure time: 2018-07-03 00:00:00
arrival time: 2018-07-03 00:00:00
departure airport: Valencia

```

Pick the flightID:

To choose a flight the customer has to enter the flightID.

Eg.

Pick the flightID: ST8909

5. Old: I did not have an option for the customer to view his/her bookings.

Changes: I have an option for that.

Snippet from the console:

```

Enter choice:
4
Enter your CustomerID:
C003
|
All your bookings are :
1. bookingID: B003
class: First
number_of_luggage: 1
booking price: 10.00
Customer_id: C003
flightID: TR9909

2. bookingID: B004
class: First
number_of_luggage: 1
booking price: 10.00
Customer_id: C003
flightID: FR1209

```

6. Old: I did not have any option for the flight administrator.
New: Now I have an option for adding and removing a flight.

CREATE TABLE AND INSERT CODE FOR FLIGHT BOOKING SYSTEM:

```
CREATE TABLE Customer (  
    CustomerID varchar(4) NOT NULL,  
    username   varchar(20) NOT NULL,  
    password   varchar(20) NOT NULL,  
    fname      varchar(15) NOT NULL,  
    lname      varchar(15) NOT NULL,  
    email       varchar(25) NOT NULL,  
    DOB        date NOT NULL,  
    phone       varchar(16) NOT NULL,  
    address     varchar(60) NOT NULL,  
    PRIMARY KEY (CustomerID));  
CREATE TABLE payment (  
    paymentID    varchar(20) NOT NULL,  
    mode_of_payment varchar(20) NOT NULL,  
    total_amount decimal(13, 2) NOT NULL,  
    CustomerID   varchar(4) NOT NULL,  
    bookingID    varchar(20) NOT NULL,  
    PRIMARY KEY (paymentID));  
CREATE TABLE booking (  
    bookingID    varchar(20) NOT NULL,  
    class        varchar(10) NOT NULL,  
    number_of_luggage smallint(6) NOT NULL,  
    booking_price decimal(13, 2) NOT NULL,  
    CustomerID   varchar(4) NOT NULL,  
    flightID     varchar(20) NOT NULL,  
    PRIMARY KEY (bookingID));  
CREATE TABLE flight (  
    flightID     varchar(20) NOT NULL,  
    flight_name   varchar(25) NOT NULL,  
    number_of_seats int(10) NOT NULL,  
    departure_date datetime NOT NULL,  
    arrival_date  datetime NOT NULL,  
    arrivalairportID varchar(20) NOT NULL,  
    departureairportID varchar(20) NOT NULL,  
    PRIMARY KEY (flightID));  
CREATE TABLE luggage (  
    luggageID    varchar(20) NOT NULL,  
    flightID     varchar(20) NOT NULL,  
    bookingID    varchar(20) NOT NULL,  
    number_of_luggage smallint(6) NOT NULL,  
    weight        decimal(13, 2) NOT NULL,  
    PRIMARY KEY (luggageID));
```

```

luggageID    varchar(5) NOT NULL,
luggage_type varchar(20) NOT NULL,
luggage_weight decimal(13, 2) NOT NULL,
PRIMARY KEY (luggageID));
CREATE TABLE airport (
  airportID   varchar(20) NOT NULL,
  airport_name varchar(25) NOT NULL,
  city        varchar(20) NOT NULL,
  country     varchar(20) NOT NULL,
  PRIMARY KEY (airportID));
CREATE TABLE booking_luggage (
  bookingID varchar(20) NOT NULL,
  luggageID varchar(5) NOT NULL,
  PRIMARY KEY (bookingID,
  luggageID));
ALTER TABLE booking ADD CONSTRAINT FKbooking67489 FOREIGN KEY (CustomerID)
REFERENCES Customer (CustomerID);
ALTER TABLE payment ADD CONSTRAINT FKpayment226222 FOREIGN KEY (bookingID)
REFERENCES booking (bookingID);
ALTER TABLE payment ADD CONSTRAINT FKpayment699130 FOREIGN KEY (CustomerID)
REFERENCES Customer (CustomerID);
ALTER TABLE flight ADD CONSTRAINT FKflight975428 FOREIGN KEY (arrivalairportID)
REFERENCES airport (airportID);
ALTER TABLE flight ADD CONSTRAINT FKflight646805 FOREIGN KEY (departureairportID)
REFERENCES airport (airportID);
ALTER TABLE booking_luggage ADD CONSTRAINT FKbooking_lu276260 FOREIGN KEY
(bookingID) REFERENCES booking (bookingID);
ALTER TABLE booking_luggage ADD CONSTRAINT FKbooking_lu755453 FOREIGN KEY
(luggageID) REFERENCES luggage (luggageID);
ALTER TABLE booking ADD CONSTRAINT FKbooking25464 FOREIGN KEY (flightID)
REFERENCES flight (flightID);
INSERT INTO Customer(CustomerID, username, password, fname, lname, email, DOB, phone,
address) VALUES ('C001', 'rose11tyler', 'badwolf10', 'rose', 'tyler', 'rose.tyler@gmail.com',
'1987-12-20', '09940-372541', 'Edenhall, Model Farm Road, Bishopstown Cork, Ireland');
INSERT INTO Customer(CustomerID, username, password, fname, lname, email, DOB, phone,
address) VALUES ('C002', 'donna22noble', 'doctordonna', 'donna', 'noble',
'donna.noble@gmail.com', '2000-11-11', '05930-377741', 'Anna Nagar, Snow Farm Road, Telly
Town, Dublin, Ireland');
INSERT INTO Customer(CustomerID, username, password, fname, lname, email, DOB, phone,
address) VALUES ('C003', 'bill33potts', 'cyberman', 'bill', 'potts', 'bill.potts@gmail.com',
'1999-10-6', '09446-311571', 'Kovan Road, Glob Centre, Abbey Town, Wales, UK');

```

```

INSERT INTO Customer(CustomerID, username, password, fname, lname, email, DOB, phone,
address) VALUES ('C004', 'river44song', 'melodypond39', 'river', 'song', 'river.song@gmail.com',
'1990-10-10', '09646-375642', 'Melborne Road, Villa Farm Road, Carmarthen, London, UK');
INSERT INTO Customer(CustomerID, username, password, fname, lname, email, DOB, phone,
address) VALUES ('C005', 'amelia55pond', 'girlwhowaited', 'amelia', 'pond',
'amelia.pond@gmail.com', '1987-12-20', '09544-245841', 'Badwolf Bay, Nomad Canter, Cape
Town Road, Cork, Ireland');
INSERT INTO airport(airportID, airport_name, city, country) VALUES ('A1', 'Cork', 'Cork',
'Ireland');
INSERT INTO airport(airportID, airport_name, city, country) VALUES ('A2', 'Heathrow',
'Longford', 'UK');
INSERT INTO airport(airportID, airport_name, city, country) VALUES ('A3', 'Chennai
International', 'Chennai', 'India');
INSERT INTO airport(airportID, airport_name, city, country) VALUES ('A4', 'Valencia', 'Valencia',
'Spain');
INSERT INTO flight(flightID, flight_name, number_of_seats, departure_date, arrival_date,
arrivalairportID, departureairportID) VALUES ('FR9896', 'Ryanair', 189, '2019-12-1', '2019-12-1',
'A1', 'A2');
INSERT INTO flight(flightID, flight_name, number_of_seats, departure_date, arrival_date,
arrivalairportID, departureairportID) VALUES ('FR6606', 'Airway', 200, '2018-10-2', '2018-10-2',
'A1', 'A3');
INSERT INTO flight(flightID, flight_name, number_of_seats, departure_date, arrival_date,
arrivalairportID, departureairportID) VALUES ('TR9909', 'AirLingus', 186, '2018-12-21',
'2018-12-22', 'A3', 'A2');
INSERT INTO flight(flightID, flight_name, number_of_seats, departure_date, arrival_date,
arrivalairportID, departureairportID) VALUES ('FR1209', 'Ryanair', 189, '2019-1-1', '2019-1-1',
'A2', 'A1');
INSERT INTO flight(flightID, flight_name, number_of_seats, departure_date, arrival_date,
arrivalairportID, departureairportID) VALUES ('ST8909', 'British Airways', 198, '2018-7-3',
'2018-7-3', 'A4', 'A1');

INSERT INTO booking(bookingID, class, number_of_luggage, booking_price, CustomerID,
flightID) VALUES ('B001', 'Economy', 1, 5.0, 'C001', 'FR9896');
INSERT INTO booking(bookingID, class, number_of_luggage, booking_price, CustomerID,
flightID) VALUES ('B002', 'Economy', 2, 5.0, 'C002', 'FR6606');
INSERT INTO booking(bookingID, class, number_of_luggage, booking_price, CustomerID,
flightID) VALUES ('B003', 'First ', 1, 10.0, 'C003', 'TR9909');
INSERT INTO booking(bookingID, class, number_of_luggage, booking_price, CustomerID,
flightID) VALUES ('B004', 'First ', 1, 10.0, 'C003', 'FR1209');
INSERT INTO booking(bookingID, class, number_of_luggage, booking_price, CustomerID,
flightID) VALUES ('B005', 'Business', 1, 8.0, 'C004', 'ST8909');
INSERT INTO booking(bookingID, class, number_of_luggage, booking_price, CustomerID,
flightID) VALUES ('B006', 'First', 1, 10.0, 'C005', 'FR9896');

```

```

INSERT INTO luggage(luggageID, luggage_type, luggage_weight) VALUES ('L001', 'Check-in',
2.3);
INSERT INTO luggage(luggageID, luggage_type, luggage_weight) VALUES ('L002', 'Carry-on',
3.1);
INSERT INTO luggage(luggageID, luggage_type, luggage_weight) VALUES ('L003', 'Check-in',
5.7);
INSERT INTO luggage(luggageID, luggage_type, luggage_weight) VALUES ('L004', 'Check-in',
2.1);
INSERT INTO luggage(luggageID, luggage_type, luggage_weight) VALUES ('L005', 'Carry-on',
2.5);
INSERT INTO luggage(luggageID, luggage_type, luggage_weight) VALUES ('L006', 'Check-in',
3.3);
INSERT INTO payment(paymentID, mode_of_payment, total_amount, CustomerID, bookingID)
VALUES ('P001', 'Visa Debit Card', 65.04, 'C001', 'B001');
INSERT INTO payment(paymentID, mode_of_payment, total_amount, CustomerID, bookingID)
VALUES ('P002', 'Visa Credit Card', 80.32, 'C002', 'B002');
INSERT INTO payment(paymentID, mode_of_payment, total_amount, CustomerID, bookingID)
VALUES ('P003', 'PayPal', 130.30, 'C003', 'B003');
INSERT INTO payment(paymentID, mode_of_payment, total_amount, CustomerID, bookingID)
VALUES ('P004', 'PayPal', 120.67, 'C003', 'B004');
INSERT INTO payment(paymentID, mode_of_payment, total_amount, CustomerID, bookingID)
VALUES ('P005', 'Visa Debit Card', 20.57, 'C004', 'B005');
INSERT INTO payment(paymentID, mode_of_payment, total_amount, CustomerID, bookingID)
VALUES ('P006', 'PayPal', 65.07, 'C005', 'B006');
INSERT INTO booking_luggage(bookingID, luggageID) VALUES ('B001', 'L001');
INSERT INTO booking_luggage(bookingID, luggageID) VALUES ('B002', 'L002');
INSERT INTO booking_luggage(bookingID, luggageID) VALUES ('B002', 'L003');
INSERT INTO booking_luggage(bookingID, luggageID) VALUES ('B003', 'L004');
INSERT INTO booking_luggage(bookingID, luggageID) VALUES ('B004', 'L004');
INSERT INTO booking_luggage(bookingID, luggageID) VALUES ('B005', 'L005');
INSERT INTO booking_luggage(bookingID, luggageID) VALUES ('B006', 'L006');

```

FLIGHT BOOKING SYSTEM TABLES GENERATED IN MYSQL OVERVIEW

Customer

CustomerID	username	password	fname	lname	email	DOB	phone	address
C001	rose11tyler	badwolf10	rose	tyler	rose.tyler@gmail.com	1987-12-20	09940-372541	Edenhall, Model Farm Road, Bishopstown Cork, Ireland
C002	donna22noble	doctordonna	donna	noble	donna.noble@gmail.com	2000-11-11	05930-377741	Anna Nagar, Snow Farm Road, Telly Town, Dublin, Ireland
C003	bill33potts	cyberman	bill	potts	bill.potts@gmail.com	1999-10-06	09446-311571	Kovan Road, Glob Centre, Abbey Town, Wales, UK
C004	river44song	melodypond39	river	song	river.song@gmail.com	1990-10-10	09646-375642	Melborne Road, Villa Farm Road, Carmarthen, London, UK
C005	amelia55pond	girlwhowaited	amelia	pond	amelia.pond@gmail.com	1987-12-20	09544-245841	Badwolf Bay, Nomad Canter, Cape Town Road, Cork, Ireland

booking

bookingID	class	number_of_luggage	booking_price	CustomerID	flightID
B001	Economy	1	5.00	C001	FR9896
B002	Economy	2	5.00	C002	FR6606
B003	First	1	10.00	C003	TR9909
B004	First	1	10.00	C003	FR1209
B005	Business	1	8.00	C004	ST8909
B006	First	1	10.00	C005	FR9896

booking_luggage

bookingID	luggageID
B001	L001
B002	L002
B002	L003
B003	L004
B004	L004
B005	L005
B006	L006

luggage

luggageID	luggage_type	luggage_weight
L001	Check-in	2.30
L002	Carry-on	3.10
L003	Check-in	5.70
L004	Check-in	2.10
L005	Carry-on	2.50
L006	Check-in	3.30

payment

paymentID	mode_of_payment	total_amount	CustomerID	bookingID
P001	Visa Debit Card	65.04	C001	B001
P002	Visa Credit Card	80.32	C002	B002
P003	PayPal	130.30	C003	B003
P004	PayPal	120.67	C003	B004
P005	Visa Debit Card	20.57	C004	B005
P006	PayPal	65.07	C005	B006

flight

flightID	flight_name	number_of_seats	departure_date	arrival_date	arrivalairportID	departureairportID
FR1209	Ryanair	189	2019-01-01 00:00:00	2019-01-01 00:00:00	A2	A1
FR6606	Airway	200	2018-10-02 00:00:00	2018-10-02 00:00:00	A1	A3
FR9896	Ryanair	189	2019-12-01 00:00:00	2019-12-01 00:00:00	A1	A2
ST8909	British Airways	198	2018-07-03 00:00:00	2018-07-03 00:00:00	A4	A1
TR9909	AirLingus	186	2018-12-21 00:00:00	2018-12-22 00:00:00	A3	A2

airport

airportID	airport_name	city	country
A1	Cork	Cork	Ireland
A2	Heathrow	Longford	UK
A3	Chennai International	Chennai	India
A4	Valencia	Valencia	Spain

IMAGES

Image number 1

Customer(BOOKING):

Enter choice: 3 Enter your CustomerID: C019 1. flightID: FR1209 flight name: Ryanair departure time: 2019-01-01 00:00:00 arrival time: 2019-01-01 00:00:00 departure airport: Heathrow 2. flightID: FR6606 flight name: Airway departure time: 2018-10-02 00:00:00 arrival time: 2018-10-02 00:00:00 departure airport: Cork 3. flightID: FR9896 flight name: Ryanair departure time: 2019-12-01 00:00:00 arrival time: 2019-12-01 00:00:00 departure airport: Cork 4. flightID: ST8909 flight name: British Airways departure time: 2018-07-03 00:00:00 arrival time: 2018-07-03 00:00:00 departure airport: Valencia 5. flightID: TR9909 flight name: AirLingus departure time: 2018-12-21 00:00:00 arrival time: 2018-12-22 00:00:00 departure airport: Chennai International Pick the flightID: TR9909	BOOKING bookingID is: B0014 Pick a class of Economy, Business or First only: Economy Number of luggages(1 or 2 only): 1 The Number or records inserted in booking is 1 Are you having luggages (y or n): y LuggageID is: L023 luggage type is(Carry-on or Check-in): Carry-on luggage weight is(kgs): 2.4 The Number or records inserted luggage is 1 The Number or records inserted in booking_luggage 1 PAYMENT PaymentID is P012 Mode of payment: Visa Debit Card, Visa Credit Card or PayPal) only: PayPal The Number or records inserted in payment is 1 BOOKING COMPLETE
--	---

Image number 2

Flight admin(add a flight)

Enter choice: 6 Enter details of the new flight Enter FlightID: A1234 Enter flight name: newflight Enter number of seats: 206 Enter departure date(yyyy-mm-dd): 2019-11-28 Enter arrival_date(yyyy-mm-dd): 2019-11-28	ARRIVAL AIRPORT INFORMATION: AirportID: A1 Airport name: Cork City:Cork CountryIreland AirportID: A2 Airport name: Heathrow City:Longford CountryUK AirportID: A3 Airport name: Chennai International City:Chennai CountryIndia AirportID: A4 Airport name: Valencia City:Valencia CountrySpain Enter arrival airport ID: A1	DEPARTURE AIRPORT INFORMATION: AirportID: A1 Airport name: Cork City:Cork CountryIreland AirportID: A2 Airport name: Heathrow City:Longford CountryUK AirportID: A3 Airport name: Chennai International City:Chennai CountryIndia AirportID: A4 Airport name: Valencia City:Valencia CountrySpain Enter departure airport ID: A2
---	--	--

Mysql for add a flight-staff. Just to show it has been added to the table

```
select *  
from flight;
```

flightID	flight_name	number_of_seats	departure_date	arrival_date	arrivalairportID	departureairportID
A1234	newflight	206	2019-11-28 00:00:00	2019-11-28 00:00:00	A1	A2

Image number 3(remove a flight-staff)

```
Enter choice:  
7  
All the flights are:  
1. flightID: A1234  
flight name: newflight  
departure time: 2019-11-28 00:00:00  
arrival time: 2019-11-28 00:00:00  
departure airport: Cork  
2. flightID: FR6606  
flight name: Airway  
departure time: 2018-10-02 00:00:00  
arrival time: 2018-10-02 00:00:00  
departure airport: Cork  
3. flightID: FR9896  
flight name: Ryanair  
departure time: 2019-12-01 00:00:00  
arrival time: 2019-12-01 00:00:00  
departure airport: Cork  
4. flightID: FR1209  
flight name: Ryanair  
departure time: 2019-01-01 00:00:00  
arrival time: 2019-01-01 00:00:00  
departure airport: Heathrow  
5. flightID: TR9909  
flight name: Airlingus  
departure time: 2018-12-21 00:00:00  
arrival time: 2018-12-22 00:00:00  
departure airport: Chennai International  
6. flightID: ST8909  
flight name: British Airways  
departure time: 2018-07-03 00:00:00  
arrival time: 2018-07-03 00:00:00  
departure airport: Valencia  
Enter the flightID you want to delete:  
A1234  
The Number or records deleted in flight is 1
```

Student statement

I hereby certify that this material which I now submit for assessment is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my own work

Signed